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Explainable AI-Based Phishing Website Detection

A Human-Centred Machine Learning and SHAP-Based Framework for Non-Technical Users

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Submission date	June 2026



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Research Overview

Field	Details
Research area	Machine learning, explainable AI, cybersecurity, human-computer interaction
Dataset	Web Page Phishing Detection (dataset_B_05_2020.csv), 11,430 instances, 87 features
Models evaluated	Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost
Best model	XGBoost — accuracy 0.96632, F1-score 0.96648, ROC-AUC 0.99443
Explainability	Global feature importance and SHAP (global and local) explanations
Interface contribution	Human-centred warning mock-up translating model output into plain language
HCI evaluation	An 8 item Likert scale with a response book. The response values listed here are synthetic demonstration data to help illustrate the operation of the workbook and do not represent empirical data on usability or trust.

Abstract

Phishing websites continue to be a dangerous cyber security threat as they mimic trusted websites to get credentials, financial information and other personal data from users. While phishing patterns can be identified with a high degree of accuracy using machine learning, a lot of phishing classifiers are black-box tools and provide little information for user, who may dismiss or be skeptical of the warning if it is not understandable. This study develops a human-centred explainable-AI framework for phishing website detection using the Web Page Phishing Detection dataset, which contains 11,430 balanced instances (5,715 legitimate and 5,715 phishing) described by 87 numeric features derived from the URL, page content and external-reputation services. Six supervised models were compared: Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting and XGBoost. Models were trained on an 80:20 stratified split and assessed with accuracy, precision, recall, F1-score and ROC-AUC, supported by five-fold stratified cross-validation. XGBoost achieved the strongest overall performance, with a test-set accuracy of 0.96632, precision of 0.96187, recall of 0.97113, F1-score of 0.96648 and ROC-AUC of 0.99443, and a cross-validated mean F1-score of 0.96906 (SD = 0.00202). Global feature importance and SHAP analysis identified external-reputation signals (notably Google indexing and PageRank) and URL/page-structure features as the most influential, and these were translated into plain-language warnings within a proposed interface mock-up. An eight-item Likert evaluation instrument and a structured response workbook were designed to support future usability testing; the response values reported here are synthetic demonstration data and do not establish usability, trust or user acceptance. The paper demonstrates that correct phishing detection must be accompanied with explanation and user-centered warning design to help users make safer decisions in terms of browsing.

Keywords: cybersecurity; explainable AI; human-computer interaction; machine learning; phishing detection; SHAP; XGBoost

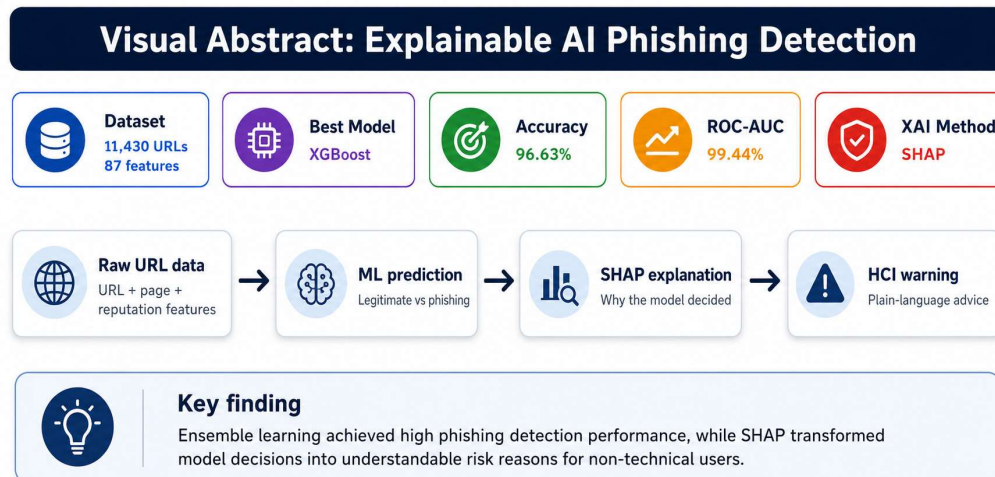


Figure 1. Visual abstract: the dataset, ensemble model, best result and SHAP-based explanation that make up the study at a glance. Source: Authors' design.

1. Introduction

Phishing is an ongoing cybersecurity issue where malicious sites mimic genuine sites and deceive users into divulging sensitive data. These sites are usually not vulnerable technically but through interface familiarity, feelings of urgency, social-engineering triggers and visual similarity. As online banking, cloud services, e-commerce and academic services continue to grow, it has become increasingly easy to create a convincing phishing page, and increasingly difficult for the average internet user to identify one.

Machine learning can detect phishing patterns across the structure of a URL, the content of a page and external-reputation signals. But if the user does not know or trust the warning from a highly accurate classifier, then the classifier will not be effective. An alert that a non-technical user receives may be ignored, not trusted, or misinterpreted, leading them to believe that they have no concern about the alert. Number of classifiers, however, are black box in nature thus restricting their use as ubiquitous decision support tools. This study fills this gap by using predictive modelling with explainable AI (XAI) and Human Computer Interaction (HCI) to communicate, in plain language, why a page is deemed risky and what the user should do, while also predicting **whether the website is a phishing site or not.**

We need to make a distinction between the four contributions of this work. The first one is predictive model development where 6 supervised models are trained and compared. The second is explainability analysis: Global feature importance and SHAP are calculated for the best model. The third one is the user-centred warning mock-up, which translates the model outputs into user-centred warnings. The fourth is the evaluation tool, an eight item Likert questionnaire and a structured response workbook are ready for future user testing. No browser extension or live detector was implemented or tested, this is only a design prototype **of the** interface.

The rest of this paper is organised as follows. Section 2 summarizes the related work on phishing detection and explainable AI and warning design. The research aim, objectives and questions are then stated in section 3. The research method is described in Section 4. Experimental results are given in Section 5. The explainability analysis is presented in section 6 as is the human-centred interface. The HCI evaluation instrument is shown in Section 7. The discussion, ethical considerations, limitations and future work, and the conclusion are provided in the following sections 8–11.

Table 1. Research problem mapping.

Current problem	Research gap	Proposed response
Phishing websites imitate trusted services and deceive users.	Detection systems often prioritise model accuracy over user understanding.	A machine-learning detector paired with user-facing explanations.
Black-box models provide limited explanation.	Non-technical users may not trust or follow warnings.	SHAP and plain-language explanations of model output.
Security warnings can be unclear or overly technical.	HCI evaluation is frequently absent from phishing-detection studies.	An HCI evaluation instrument and a structured response workbook.

Phishing Threat and Explainable-AI Defense Flow

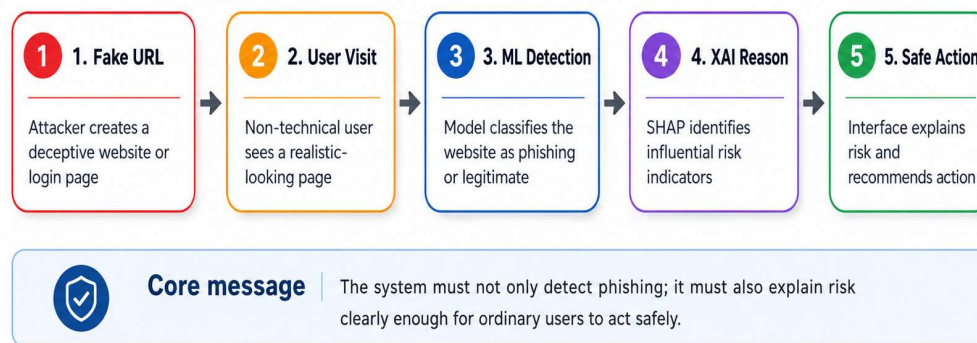


Figure 2. Phishing threat and explainable-AI defence flow, from a deceptive URL through model detection and SHAP-based reasoning to a clear user warning. Source: Authors' design.

2. Literature Review

2.1 Phishing Website Detection Approaches

There are a number of complementary strategies that have been developed to detect phishing. Blacklists are lists of known malicious URLs that are fast and accurate, but cannot recognise new

domains, due to the time periods of many phishing sites. Heuristic and rule-based systems represent knowledge by a set of hand-crafted rules, are easily interpretable but brittle and need frequent manual changes. URL-based detection checks the length of the URL, unusual characters in the URL, host properties of the URL, and IP address detection. In Content-Based detection, forms, hyperlinks, embedded scripts, etc. and even the login fields of the page are studied. Reputation-based detection relies on metrics like web traffic and domain age, search-engine indexing. Visual-similarity methods compare the photo-realistic representation of a candidate page to that of authentic brands. More recent work uses machine-learning and deep-learning approaches, which learn patterns directly from large features sets, with a better ability to generalize to new pages that were not seen before than static rules or lists [1, 2]

2.2 Machine Learning for Phishing Detection

Supervised classifiers have been widely applied to the problem of phishing detection on structured-feature sets. Logistic Regression offers a linear baseline, which is easily interpreted, but is unable to account for complex non-linear interactions. A single Decision Tree's information is easy to understand, but could be over-fitting. There Ensemble methods exist to overcome the weakness: Random Forest is based on the idea of averaging many decorrelated Trees and Extra Trees brings more randomisation for robustness on tabular data. Gradient Boosting adds trees one after another to minimise previous errors while Neural-network methods also work well when there is a lot of raw data (character-level URLs, page screenshots), but are not as competitive as gradient-boosted trees on moderate-sized tabular feature sets, and they are less interpretable. The features in phishing are heterogeneous, the relationships between different features are non-linear, and the interactions between features are non-linear and there's no support of axis-aligned splits for them, hence the use of tree-based ensembles works well here [2, 5].

2.3 Explainable AI in Cybersecurity

An accurate but opaque model offers little support for human security decisions. Explainable AI enables you to differentiate global explanations, which explain overall model behavior, from local explanations, which explain the reasoning behind a given prediction. Feature-importance scores assign a rank to the inputs based on the overall contribution. SHAP gives each feature its contribution to a particular prediction with the help of Shapley values in the context of cooperative game theory, giving a consistent and locally accurate explanation of the reason a page was classified as phishing or legitimate [3]. LIME, on the other hand, provides a complementary approach by simplifying the model in the neighborhood of a specific instance to approximate it [4]. The advantages of SHAP are that it is based on a sound theoretical framework, provides additive local explanations and informative visualisations for summarising; its disadvantages are that it is computationally expensive and that feature attributions can be interpreted as causal effects. When it comes to security decision support, explanations need to be faithful to the model, but also easy enough for a non-expert to follow and implement.

2.4 Human-Computer Interaction and Warning Design

The effectiveness of a security warning is found to rely not only on the accuracy of the detector underpinning it, but also on the design of the warning – HCI research reveals. Warnings should be understandable, include good risk communication and be actionable [6] [7]. If the warnings are repeated or of poor quality, they can lead to warning fatigue and habituation, whereby the user no longer pays attention to the warning. Good designs may avoid jargon and use colour and visual hierarchy to emphasise severity (in a way that is understandable to people with colour vision defects) and clearly define what the user should do next. A key objective is proper trust calibration: If the system is likely to be correct, the user should do so, and if it is not, then the user should be careful, but not take it for granted. Research on the reasons for phishing's success focuses on the need for non-technical users to be told about the risks they can see, not on technical details [8].

2.5 Research Gap

The phishing-detection literature tends to focus on predictive performance and neglect factors of human understanding, explanation quality, warning usability, trust calibration and decision support. A few studies have combined the three fields into a single end-to-end pipeline for the user, but not many. This study tackles that issue by integrating a good performing classifier with SHAP based explanations and warning design based on the human factor; and by creating a structured instrument that can be used for empirical evaluation of usability, explanation clarity, trust and decision support in future.

3. Research Aim, Objectives and Questions

The aim of this study is to design and assess an explainable-AI framework for phishing website detection that accurately classifies websites and presents understandable, actionable warnings to non-technical users.

Table 2. Research aim, objectives and research questions.

Objective	Related research question
01. Pre-process and analyse the Web Page Phishing Detection dataset.	RQ1. What are the characteristics and class balance of the dataset?
02. Compare six supervised models for phishing classification.	RQ2. Which model achieves the strongest overall performance?
03. Identify the most influential URL, page-content and external-service features.	RQ3. Which features contribute most to phishing predictions?
04. Apply SHAP to explain global and individual predictions.	RQ4. How can explainable AI improve the transparency of the detector?
05. Design a plain-language warning interface mock-up for non-technical users.	RQ5. How can model output be presented to non-technical users?
06. Develop an HCI evaluation instrument for	RQ6. How can usability, explanation clarity,

Objective	Related research question
usability, clarity, trust and decision support.	trust and decision support be measured?

4. Research Methodology

4.1 Research Design

The study uses a quantitative, experimental machine-learning design combined with an explainable-AI analysis and a human-centred interface-design component. The modelling and explainability work is empirical and reproducible from the dataset and notebook; the interface is a design prototype, and the accompanying evaluation instrument is prepared for future user testing.

4.2 Dataset Source and Characteristics

The Web Page Phishing Detection dataset (file dataset_B_05_2020.csv) was used. Published on Mendeley Data (version 3) by Hannousse and Yahiouche, it is related to an experimental study of a set of benchmark data for website phishing detection [1, 2]. The raw file has 11,430 website records and 89 columns: 11,430 columns identified by the identifier/target column, and one column containing non-numeric URL text which was excluded from the feature matrix; after removing the non-numeric URL text column and the identifier/target columns, 87 numeric features remain in the feature matrix. The dataset is balanced with 5715 legitimate and 5715 phishing websites and the target variable is binary 'status'. There are three types of features: URL-related, page-content and external-service signals (Table 4). One caveat is that the data here is reflective of the time the data was collected, and may not reflect the latest phishing practices (see Section 10 for more discussion).

Table 3. Dataset characteristics.

Dataset attribute	Value
Dataset name	Web Page Phishing Detection
File used	dataset_B_05_2020.csv
Source / version	Hannousse & Yahiouche, Mendeley Data, V3 (DOI: 10.17632/c2gw7fy2j4.3)
Raw shape	11,430 rows × 89 columns
Modelling features	87 numeric features
Total instances	11,430
Legitimate websites	5,715
Phishing websites	5,715
Target variable	status (0 = legitimate, 1 = phishing)

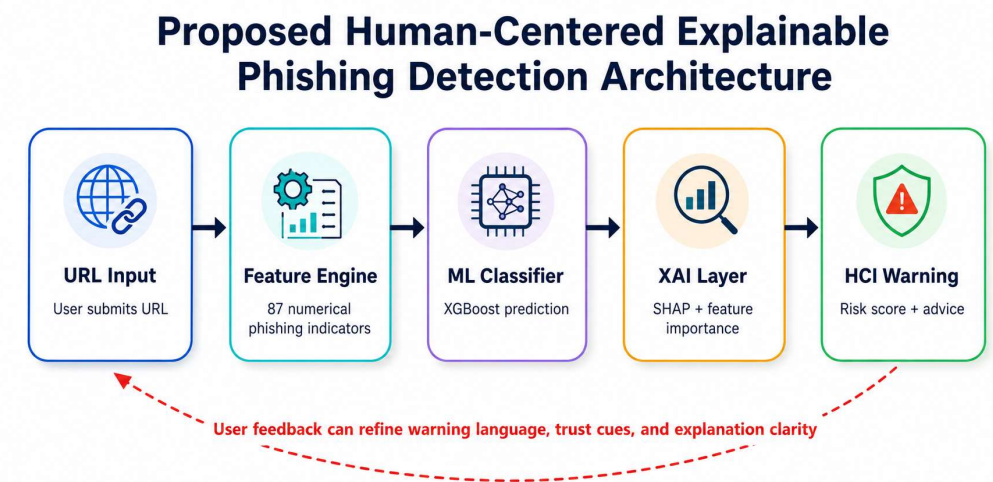


Figure 3. Proposed human-centred explainable phishing-detection framework, integrating the dataset, modelling pipeline, explainability layer and interface output. Source: Authors' design.

Feature Taxonomy Used in the Dataset



Figure 4. Dataset feature taxonomy: URL-based, page-content and external-service features. Source: Authors' analysis.

Table 4. Feature categories used in the dataset.

Feature category	Examples	Explanation potential for users
URL-based features	length_url, nb_dots, nb_hyphens, nb_qm, ratio_digits_host	Suspicious URL length, unusual symbols or confusing structure.
Page-content features	nb_hyperlinks, login_form, iframe, safe_anchor	Risky page behaviour and abnormal linking patterns.
External-service features	google_index, page_rank, web_traffic, dns_record, domain_age	Weak reputation, missing index signals or low domain trust.

Balanced Dataset Class Distribution

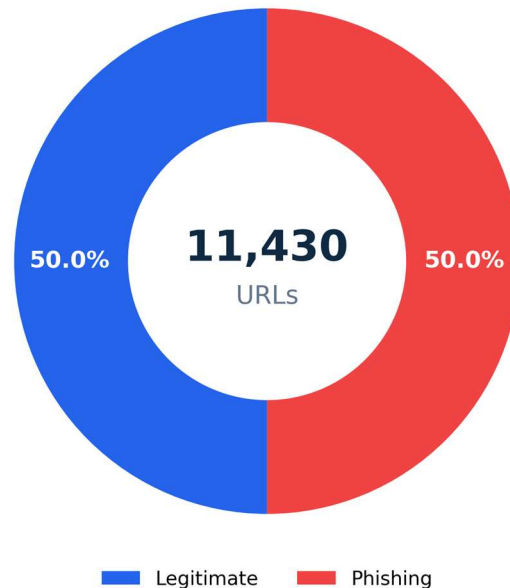


Figure 5. Balanced class distribution of the Web Page Phishing Detection dataset. Source: Authors' analysis.

4.3 Data Preprocessing

Using pandas, the data was loaded into Python. The status of the target variable was created by assigning 0 for legitimate and 1 for phishing. The raw URL text field was eliminated from the modelling matrix as it is a free text identifier and could contain instance-specific information if it remained. The data type and missing value checks and the invalid value checks were performed on the columns, as this dataset was already pre-extracted as a numeric feature of the specific data and did not need to be imputed during this analysis. The data was split into an 80–20 train–test set with stratification to ensure a balanced class distribution in each set (1,143 legitimate and 1,143 phishing instances in the test set). A fixed random seed was set, to enable the split to be reproducible.

Researcher confirmation required: confirm the exact treatment of any missing/duplicate records, whether feature scaling was applied (and, if so, that it was fitted inside a pipeline on the training folds only), and the specific random-seed value used for the split and models.

4.4 Model Development

To implement the supervised classifiers, the scikit-learn and XGBoost library was used, with the following classifiers implemented: Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting and XGBoost classifier. In this study, no systematic hyperparameter search (e.g., grid or randomised search) was carried out and models were trained with the default hyperparameters found in the provided library. The reported results are explicitly given as limitation, as it may be possible to achieve even better performance after tuning. The best focus for

the optimisation of model selection was the F1 Score for the phishing class, which showed the penalties associated with missed detections.

Hyperparameters to confirm: If they are present, confirm exactly what they are and if class-weighting was used, whether this was applied or not and if tuned, whether it was tuned for each model.

4.5 Cross-Validation

Five-fold stratified cross validation was used to evaluate the stability of the model with the F1-score of phishing class as the evaluation metric. The results of the cross validation are tabulated as mean and standard deviation over the folds (Table 7). Any information-dependent preprocessing should be fit into each training fold, and not on the entire data, to prevent leakage.

4.6 Evaluation Metrics

The metrics used to measure performance are as follows: those as defined in table 9. Of particular importance in phishing detection are recall and the false-negative rate: a false-negative (a phishing page that is not detected) can result in a user being vulnerable to credential theft, while a false-positive (a correct identification of a phishing page) mainly means a nuisance and, if frequent, "warning fatigue. The two types of error are thus reported. In case of the precision-recall analysis, the average precision (AP) is reported rather than the ROC-AUC and the latter should not be confused with the former.

Table 9. Evaluation-metric definitions.

Metric	Definition
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision	$TP / (TP + FP)$
Recall (sensitivity)	$TP / (TP + FN)$
F1-score	$2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$
Specificity	$TN / (TN + FP)$
False-positive rate	$FP / (FP + TN)$
False-negative rate	$FN / (FN + TP)$
ROC-AUC	Area under the receiver-operating-characteristic curve
Average precision (AP)	Summary of the precision-recall curve

4.7 Explainability Procedure

The top performing model (XGBoost) was chosen for explanation. We employed two complementary techniques: The built-in feature importance of the model for global ranking and the SHAP global and local explanations. A positive SHAP value pushes a prediction towards the phishing class, a negative value pushes a prediction towards the legitimate class. Global explanations provide a summary of feature effects over multiple instances (SHAP summary plot); local explanations will explain why a specific decision was made. These technical attributions were

then translated to plain-language statements (e.g., a strong negative contribution from `google_index` becomes “the website has a poor search-engine reputation”) which were presented on the interface.

Yes: confirm the type of explainer (TreeExplainer), background/reference data, and the sample size used to compute the SHAP.

4.8 Interface Design

The interface is intended for non-technical users making everyday browsing decisions. The design requirements were: present a clear prediction label and a numeric risk score; provide a short, plain-language explanation of the main reasons; offer concrete safety advice; use colour-coded risk indicators supported by text (so meaning does not depend on colour alone); avoid technical jargon; and present a clear warning hierarchy. To support trust calibration, the interface communicates the basis for the warning rather than asserting certainty, so that users can judge how far to rely on it. The interface is a static mock-up (Figure 16) and was not implemented as working software.

4.9 Software and Reproducibility

The experiments were conducted in Python using pandas, NumPy, scikit-learn, XGBoost, SHAP and Matplotlib. Exact library versions are not recorded in the available materials and are left as placeholders below to be completed by the researcher; they are not invented here.

- Python
- pandas / NumPy
- scikit-learn
- XGBoost
- SHAP
- Matplotlib

The trained model and result files (`best_phishing_detection_model.pkl`, `model_performance_results.csv`, `cross_validation_results.csv`, `feature_importance_results.csv`, and the figure set) are available with the project and support reproduction of the reported results.

5. Results and Statistical Analysis

5.1 Model Performance

The six classifiers were compared on the held-out test set (Table 6). XGBoost achieved the strongest overall performance, with an accuracy of 0.96632, precision of 0.96187, recall of 0.97113, F1-score of 0.96648 and ROC-AUC of 0.99443. Extra Trees and Random Forest followed closely, confirming that tree-based ensembles are well suited to this structured dataset, while Logistic Regression and a single Decision Tree were the weakest. The high recall of XGBoost is desirable here because it corresponds to few missed phishing pages.

Table 6. Model performance comparison on the held-out test set.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
XGBoost	0.96632	0.96187	0.97113	0.96648	0.99443
Extra Trees	0.96457	0.96416	0.96500	0.96458	0.99298
Random Forest	0.96194	0.95833	0.96588	0.96209	0.99355
Gradient Boosting	0.95319	0.94965	0.95713	0.95338	0.99049
Logistic Regression	0.93613	0.93921	0.93263	0.93591	0.98142
Decision Tree	0.93395	0.92907	0.93963	0.93432	0.93395

Note: precision, recall and F1-score are reported for the phishing class. All values are rounded to five decimal places.

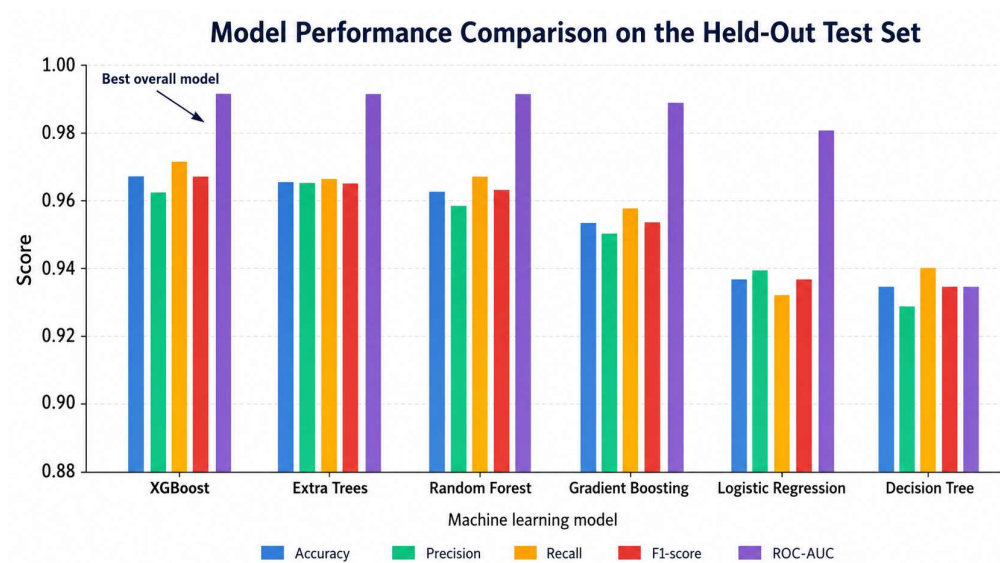


Figure 6. Model performance comparison across all metrics (accuracy, precision, recall, F1-score and ROC-AUC) on the held-out test set, with XGBoost as the best overall model. Source: Authors' analysis.

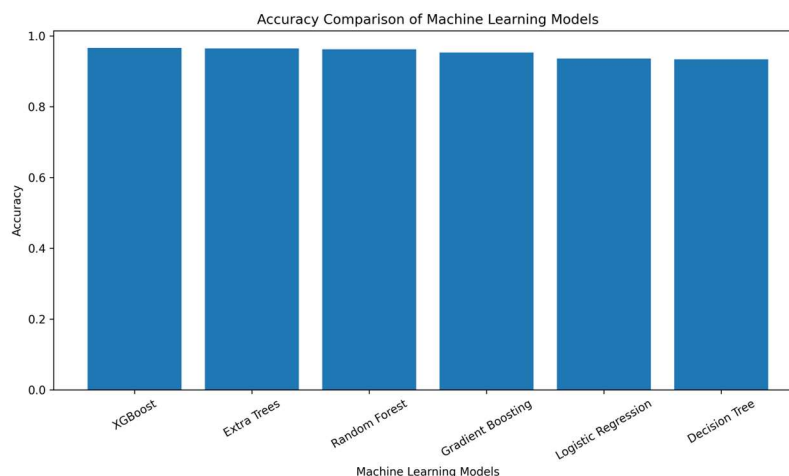


Figure 7. Model accuracy comparison. Source: Authors' analysis.

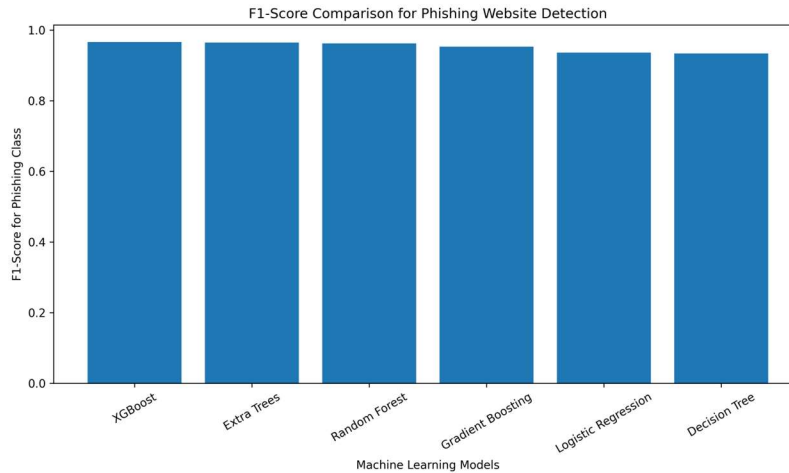


Figure 8. Model F1-score comparison. Source: Authors' analysis.

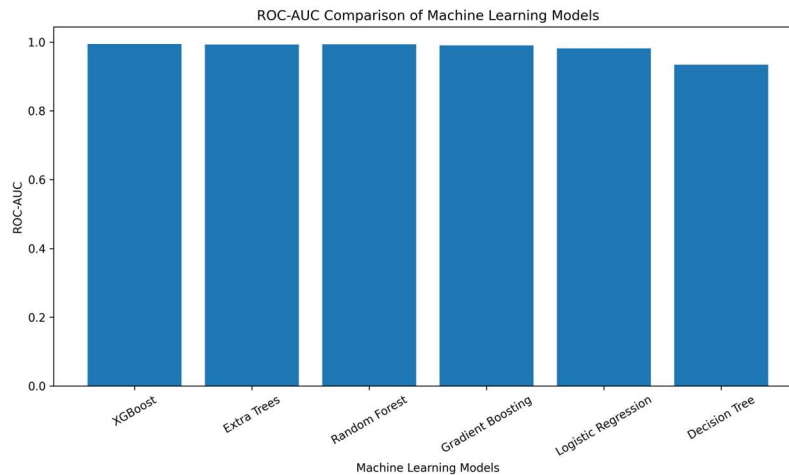


Figure 9. Model ROC-AUC comparison. Source: Authors' analysis.

5.2 Cross-Validation Stability

Five-fold stratified cross-validation confirmed that the leading models are stable across folds (Table 7). XGBoost achieved a mean cross-validated F1-score of 0.96906 with a standard deviation of 0.00202, the highest mean and the smallest variability of the three top models, indicating consistent performance rather than a result that depends on a single fortunate split.

Table 7. Cross-validation statistics for the top models.

Model	Mean CV F1-score	SD of CV F1-score
XGBoost	0.96906	0.00202
Random Forest	0.96705	0.00461
Extra Trees	0.96341	0.00455

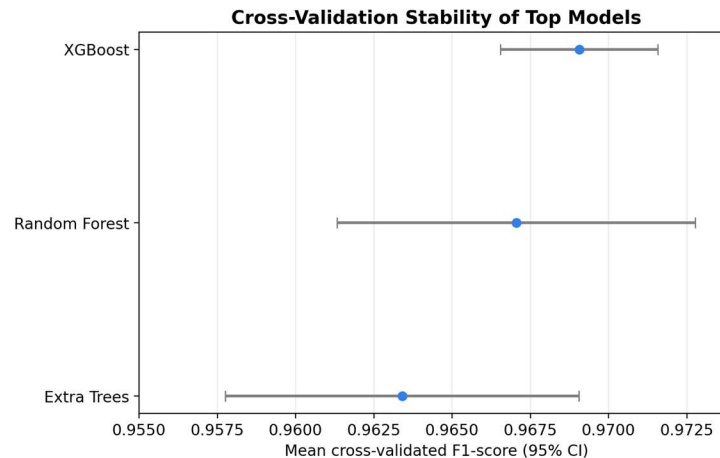


Figure 10. Cross-validation stability of the top models (mean F1-score with standard deviation). Source: Authors' analysis.

5.3 Confusion-Matrix Analysis

The results showed that XGBoost classified 1,099 legitimate pages and 1,110 phishing pages correctly on test set (2,286 pages) where false positives were 44 pages and false negatives were 33 pages (Figure 11). The statistics in Table 8 are internally consistent with the headline metrics in Table 6: accuracy, precision, recall and F1-score are all integral parts of these four counts. The most important value for security is the 2.89% false negative rate rate, meaning that this is the number of phishing pages that the model will miss; the false positive rate of 3.85% is less harmful but if it happens often, it can lead to warning fatigue.

Table 8. Confusion-matrix-derived statistics for the XGBoost model.

Statistic	Value	Interpretation
True negatives	1,099	Legitimate websites correctly classified.
False positives	44	Legitimate websites incorrectly flagged as phishing.
False negatives	33	Phishing websites missed by the model.
True positives	1,110	Phishing websites correctly detected.
Specificity	0.96150	Proportion of legitimate sites correctly identified.
False-positive rate	3.85%	Potential warning-fatigue risk.
False-negative rate	2.89%	Potential security-exposure risk.

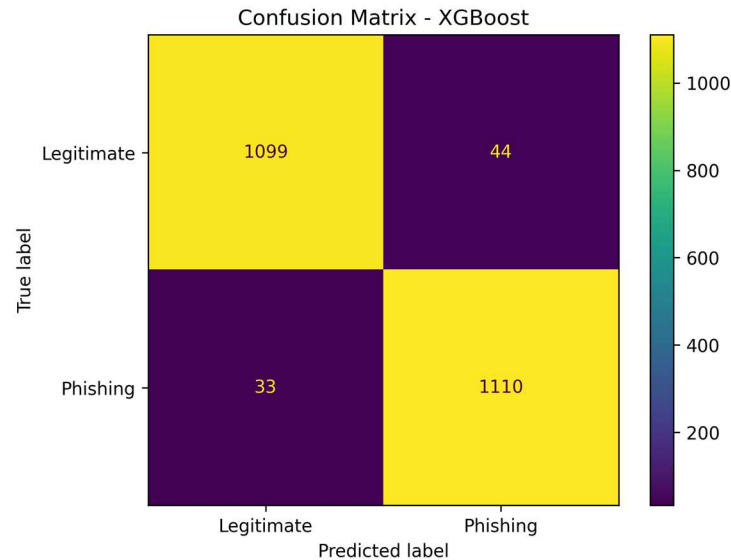


Figure 11. Confusion matrix for the XGBoost model on the test set. Source: Authors' analysis.

5.4 ROC and Precision-Recall Analysis

The ROC curve (Figure 12) demonstrates that the model is separating between the two classes well across thresholds and is a good fit with the ROC AUC of 0.99443. The precision-recall curve (Figure 13) summarises the performance for the phishing (positive) class, and reports an average precision of around 0.99. The dataset is balanced, so the ROC and precision-recall analysis essentially tell the same story here, although the average precision is a different number than the ROC-AUC, and is reported separately.

The researcher has to confirm, write down the actual value of the average-precision from the notebook (rounded to two decimal places)..

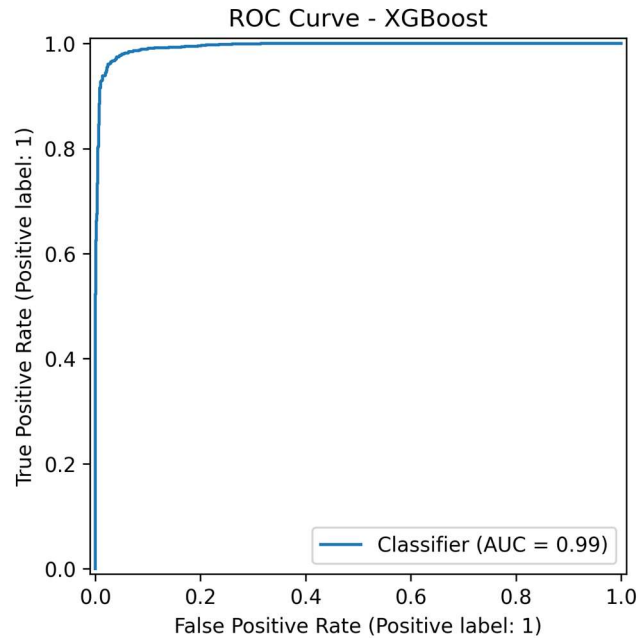


Figure 12. ROC curve for the XGBoost model ($AUC \approx 0.99$). Source: Authors' analysis.

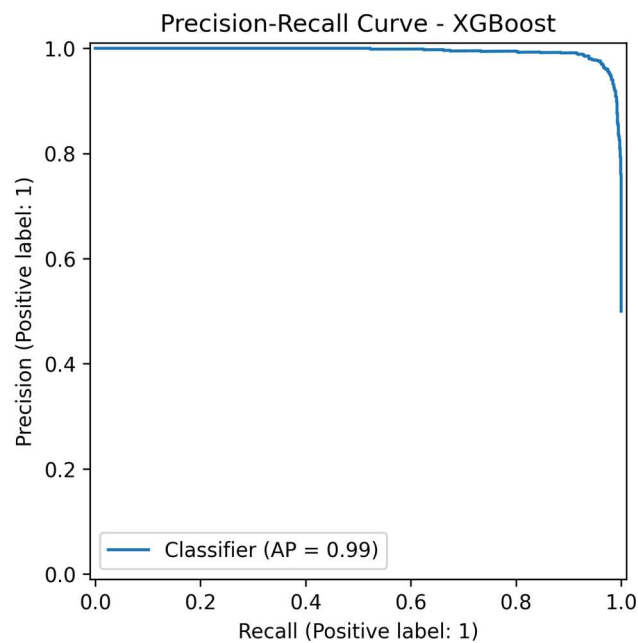


Figure 13. Precision-recall curve for the XGBoost model (average precision ≈ 0.99). Source: Authors' analysis.

6. Explainable AI and Human-Centred Interface

6.1 Feature Importance

The gain-based feature importance of the XGBoost model (Figure 14) is dominated by `google_index`, followed at a much lower level by `page_rank`, `nb_qm`, `nb_www`, `nb_hyperlinks`,

ratio_digits_host, dns_record, phish_hints, suspicious_tld and length_words_raw. The prominence of external-reputation signals (Google indexing, PageRank, DNS records) alongside URL and page-structure features indicates that the model relies on a combination of how reputable a domain appears and how its address and content are constructed.

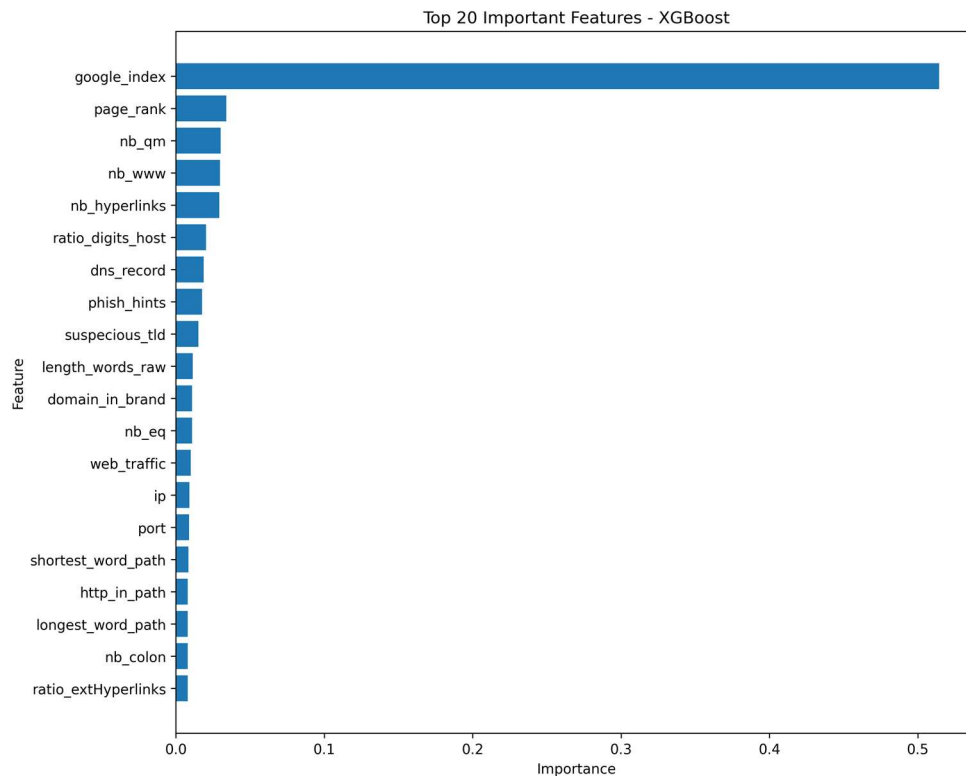


Figure 14. Top 20 feature-importance scores for the XGBoost model. Source: Authors' analysis.

6.2 SHAP-Based Explainability

An analysis of individual features through SHAP (Figure 15) explains and ranks the influence of each feature on predictions. Things such as if they have a high or low google_index for phishing, or a low page_rank for legitimacy. They are translated into inferences using natural language processor interface warnings, mappings from the internal logic of the model to the user.

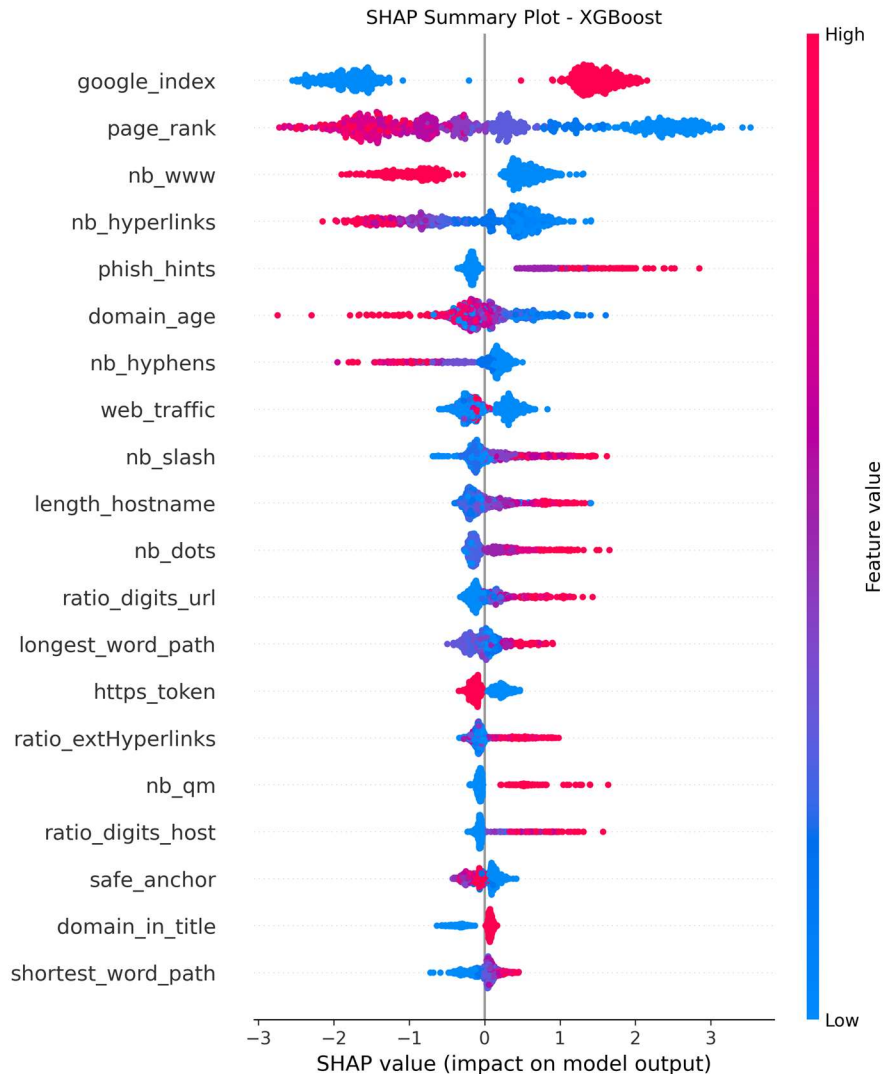


Figure 15. SHAP summary plot for the XGBoost model. Source: Authors' analysis.

6.3 Human-Centred Interface Mock-Up

The newly proposed user interface is carefully designed to communicate only four specific, highly visible components in particular (an explicit prediction label, a numerical risk score that is calibrated, a clear plain-language explanation and concrete actionable safety advice; see Fig. 16). This interface uses an intuitive, colour-coded risk indicators backed by explanatory text labels and a carefully curated warning hierarchy while consciously omitting confusing technical jargon for improved clarity, better understanding, and accessibility. In some cases in which the detected risk is considered as very high there provided safety hints firmly guide users with direct instructions not to enter any sensitive passwords, banking credentials or private personal data and strongly recommending them attempt instead to reach the targeted application safely only through its confirmed official communication channels. This image shown in the figure is only a static version of a concept design mock-up and not yet an actual code that has worked into software or used for daily consumer use as part of an integrated, browser plug-in.

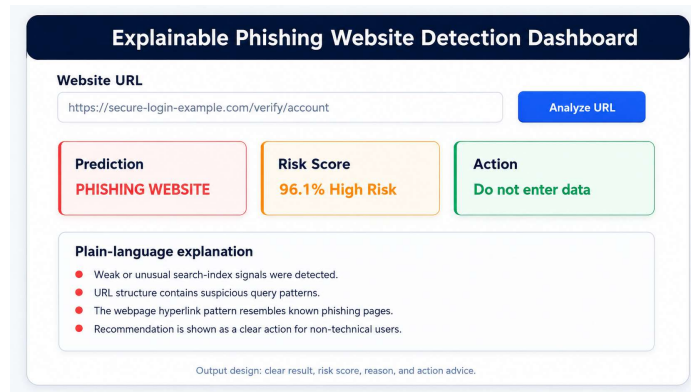


Figure 16. Human-centred warning-interface mock-up for non-technical users. Source: Authors' design.

6.4 HCI Evaluation Instrument

An eight-item Likert instrument (Table 10) was designed to assess usability, warning clarity, explanation clarity, risk communication, visual-cue effectiveness, decision support, trust and suitability for non-technical users, each on a five-point scale from “strongly disagree” to “strongly agree”. An accompanying response workbook records participant identifiers, demographic fields, individual item responses and summary statistics, and can be reused to record and analyse responses in a future study.

Table 10. HCI evaluation instrument.

Item	Questionnaire item	Construct
Q1	The system was easy to use.	Usability
Q2	The phishing warning was easy to understand.	Warning clarity
Q3	The explanation helped me understand why the website was risky.	Explanation clarity
Q4	The risk score was useful for judging the website.	Risk communication
Q5	The colour-coded warning helped me identify risk quickly.	Visual-cue effectiveness
Q6	The safety advice helped me make a safer decision.	Decision support
Q7	I would trust this system before entering personal data.	Trust
Q8	The interface is suitable for non-technical users.	User fit

7. Demonstration of the HCI Evaluation Instrument

The data presented here are synthetic demonstration data created to show how the evaluation workbook records responses and calculates summary statistics. These are not actual responses from participants and do not provide any insight into the usability, clarity, trustworthiness, effectiveness, or user acceptance of the interface. No conclusions about real user behavior can be drawn from these data. Detailed synthetic tables and charts are included in Appendix A.

Researcher confirmation is required: verify whether any responses were collected from real participants or if they are solely for demonstration. If real data exist, a comprehensive HCI methodology section is needed before presenting these values as empirical evidence. This should include details on design, recruitment, sampling, inclusion/exclusion criteria, demographics, testing environment, tasks, consent, anonymity, analysis, and any ethical approvals or exemptions.

For illustration, the workbook includes 30 synthetic response sets across eight items, with an overall mean score of 4.26 out of 5, an average agreement rate of 94.6%, and a Cronbach's alpha of 0.895. Strictly as a demonstration, these figures show that the workbook accurately computes item means, agreement rates, and internal consistency; they do not indicate how real users might rate the interface. The per-item summaries and distributions are shown in Appendix A (Tables 11–12 and Figures 14–15).

8. Discussion

This research was focused on creating an explainable, human-centred phishing detection system and assessing its parts with respect to research questions found in Table 2.

Regarding RQ1 - RQ2, the database is balanced across 11,430 instances and XGBoost has performed much better than any of the other 6 models. It is believed that the reason XGBoost has performed so well was due to the fact that gradient boosted trees with regularisation capture the non-linear relationships of the heterogeneous page features, URLs and reputation features, and they are able to control for over-fitting. Further, the fact that Extra Trees and Random Forest achieved performance levels very close to that of the XGBoost supports the generalisation from previous studies that tree ensembles will be successful for working on structured problems, in a similar way to previous benchmark studies performed on the same dataset as this study [2].

Regarding RQ3 - RQ4, External Reputation Features, especially Google indexing and PageRank were identified by both Feature Importance and SHAP as being the most influential external-repute features. The high level of reliance on Google indexing is of operational significance, as it is highly discriminative however will require an external service to provide service in order to be able to use it to produce the necessary information. In addition, both Feature Importance and SHAP ranked the URL and Page Structure Signals; those signals include: number of characters in the URL, presence of query string, presence of www, pattern of hyperlinks, ratio of digits to letters in the URL and DNS Records.

The information provided by SHAP complements the ranking of the individual page feature values with respect to the influence of that page feature for that particular page. Thus, SHAP provides an explanation to help non-technical audiences understand the specific nature of each page feature when making a phishing detection.

Also, the findings show that interpretability and accuracy have a relationship with one another. In this instance, having a low false negative rate is of particular importance since missing a phishing page could directly lead to a user losing their credentials. The main impact of false positives will result in annoying users. With enough false positives, users become accustomed to them, leading to

a level of “warning fatigue.” The user interface proposed ties SHAP (Shapley Additive Explanations) explanation interpretable models to user interface warnings so that every time a prediction is generated, the user sees the top reasons for the prediction and the suggested action. The design is therefore aimed at providing trust calibration, rather than providing absolute trust in the warning; the user is expected to trust the warning provided that there is supporting evidence and will approach the warning cautiously otherwise. Usable and secure are different attributes; having a “user-friendly” user interface will not necessarily result in “user-secure” behaviour; thus user testing must be conducted to collect empirical data.

The key theoretical contribution is an integrated pipeline that links the prediction, explanation, and user interface design for laypeople. The main practical contribution includes a reproducible modelling notebook, an explanation layer(s), wireframe mockup(s), and a reusable evaluation tool. Because the responses to the HCI questions are synthetic, the discussion will only assert that the evaluation framework is ready for future use, not that the trust or usability has been demonstrated.

9. Ethical Considerations

This study presents a number of ethical issues that are summarised in Table 13. Although public access to the phishing dataset has been established, and public use has been restricted to protection-related research, there is still an obligation to treat the data with care, and only to use it to produce benefit. All malicious URLs had been extracted in advance and only their numeric form was used; no phishing website was visited or interacted with during the completion of this study.

Any future work involving actual participants requires that the following criteria are met:

1. Voluntary, informed consent must be provided;
2. Participants must be allowed to withdraw at any time;
3. Any responses must be anonymised;
4. Data must be stored securely to protect the confidentiality of participants;
5. A documented ethical approval or an ethical waiver must be obtained; no such waiver has been provided for this research project.

The design of any interface should take into account accessibility for users who are colour blind.

Automation bias and over-reliance on model predictions are potential operational risks. A false positive could damage the reputation of a legitimate site, while a false negative could expose a user to attack. Consequently, warnings issued from these models should communicate uncertainty rather than certainty.

As part of the ethical posture of this study, the ethical use of data, consideration of dataset bias, and accountability surrounding the creation of synthetic demonstration data must be recognised through ethical disclosure..

Table 13. Ethical and reproducibility considerations.

Area	Consideration
Dataset ethics	The dataset is public and used only for defensive academic research; public availability does not remove ethical responsibility.
Safe handling	Malicious URLs were used only as numeric features; no live phishing site was accessed.
User evaluation	A real HCI study requires informed consent, voluntary participation, the right to withdraw and anonymisation.
Synthetic data	The reported HCI responses are synthetic demonstration data and must not be presented as real participant data.
Automation bias	Warnings should support, not replace, user judgement and should communicate uncertainty.
Error consequences	False positives risk reputational harm; false negatives risk security exposure.
Reproducibility	Dataset version, metrics, notebook and output files are documented for replication.

10. Limitations and Future Work

There are a number of constraints that should be taken into account. The model, which is trained and validated on one publicly available data set, may be old and may not account for the full range of the phishing websites discovered at this time. There was no external validation performed using any independent data sets nor using live feeds. The features of this model are extracted statically, so the model would not adapt to the evolution of phishing tactics or respond to concept drift, and it has not been tested for adversarial-robustness. The features of this detector depend on external services (Google Index, PageRank, DNS and web traffic); therefore, they cannot guarantee their availability or the latest data. This model has been set up with default values for its hyperparameters and has not been extensively evaluated for how feature interactions occur. The computation of SHAP is computationally expensive when computed at the scale described in this work. The complete interface to the tool is a mockup, as opposed to an actual browser plug-in. In addition, the data used for the evaluation of human-computer interaction on the interface is synthetic with no participants involved (e.g., no demographic diversity) and with only subjective measures; colour vision accessibility of the evaluator has not been evaluated through formal methods.

Future work should validate the model on external datasets and real-time phishing feeds; implement and deploy a browser-extension prototype; add adversarial testing, concept-drift monitoring and periodic retraining; and extend explanations to multiple languages, QR-code phishing and screenshot-based visual-similarity analysis. On the HCI side, future work should conduct a properly designed user study with real participants, task-based usability measures, a validated usability scale (such as the System Usability Scale) administered in practice, mobile-

interface and accessibility testing, longitudinal studies of trust calibration, and a controlled comparison of ordinary warnings against explanation-based warnings.

11. Conclusion

This study aimed to develop an explainable, human-centred framework for phishing website detection that pairs accurate classification with understandable warnings for non-technical users. Using a balanced dataset of 11,430 websites described by 87 features, six supervised models were compared under an 80:20 stratified split with five-fold cross-validation. XGBoost achieved the strongest overall performance, with a test-set accuracy of 0.96632, an F1-score of 0.96648 and a ROC-AUC of 0.99443, and a cross-validated mean F1-score of 0.96906. Feature importance and SHAP analysis showed that external-reputation signals and URL/page-structure features drive the predictions, and these were translated into plain-language warnings within an interface mock-up. An eight-item evaluation instrument and a structured response workbook were prepared to support future usability testing; the response values reported here are synthetic demonstration data and do not establish usability, trust or user acceptance. Overall, the work supports the view that effective phishing defence should combine accuracy, explainability and user-centred warning design, while recognising that empirical evaluation with real users is needed before claims about usability and trust can be made.

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Appendix A. Synthetic HCI Demonstration Data

The tables and figures in this appendix present **synthetic demonstration data** only. They illustrate the operation of the evaluation workbook and must not be interpreted as evidence about real users.

Table 11. Illustrative HCI summary statistics (synthetic demonstration data).

Item	Mean	SD	Agreement	Construct
Q1	4.27	0.58	93.3%	Usability
Q2	4.33	0.48	100.0%	Warning clarity
Q3	4.27	0.58	93.3%	Explanation clarity
Q4	4.10	0.61	86.7%	Risk communication
Q5	4.57	0.50	100.0%	Visual-cue effectiveness
Q6	4.40	0.56	96.7%	Decision support
Q7	3.90	0.40	86.7%	Trust
Q8	4.27	0.45	100.0%	User fit

Note: synthetic values. The highest-rated item is Q5 (visual-cue effectiveness) and the lowest is Q7 (trust before entering personal data). These patterns are illustrative only.

Table 12. Illustrative Likert response distribution (synthetic demonstration data).

Item	Strongly disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly agree (5)
Q1	0	0	2	18	10
Q2	0	0	0	20	10
Q3	0	0	2	18	10
Q4	0	0	4	19	7
Q5	0	0	0	13	17
Q6	0	0	1	16	13
Q7	0	0	4	25	1
Q8	0	0	0	22	8



Figure 17. HCI evaluation summary dashboard: overall mean, agreement, internal consistency and highest-rated item (synthetic demonstration data). Source: Authors' demonstration.

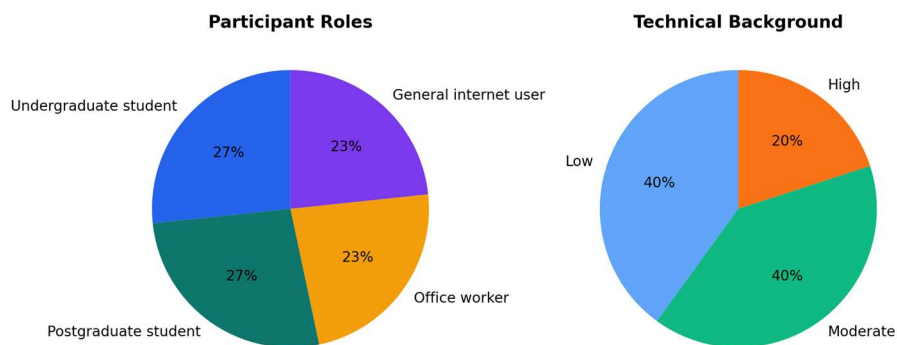


Figure 18. Illustrative participant profile by role and technical background (synthetic demonstration data). Source: Authors' demonstration.

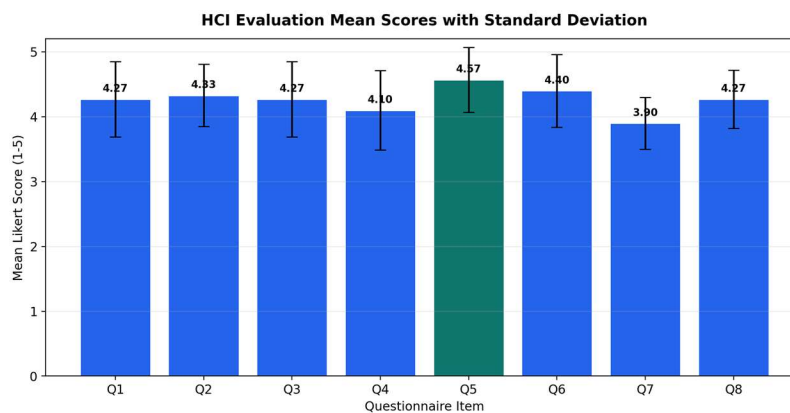


Figure 19. Illustrative item means with standard deviation (synthetic demonstration data). Source: Authors' demonstration.

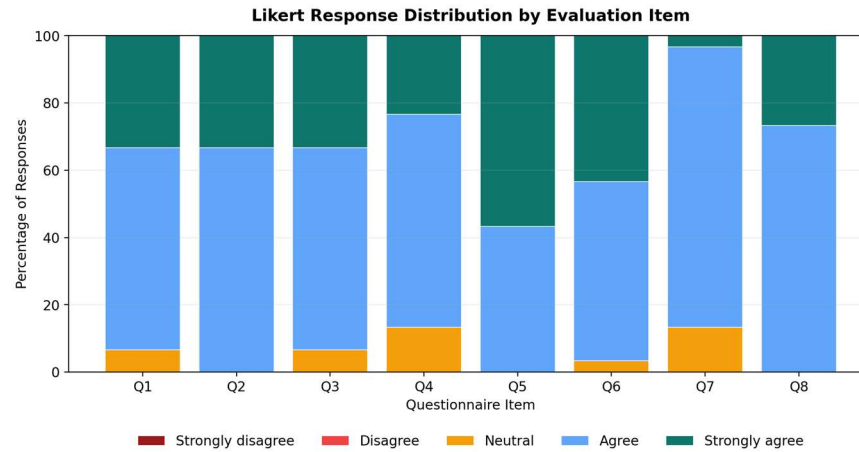


Figure 20. Illustrative Likert response distribution by item (synthetic demonstration data). Source: Authors' demonstration.

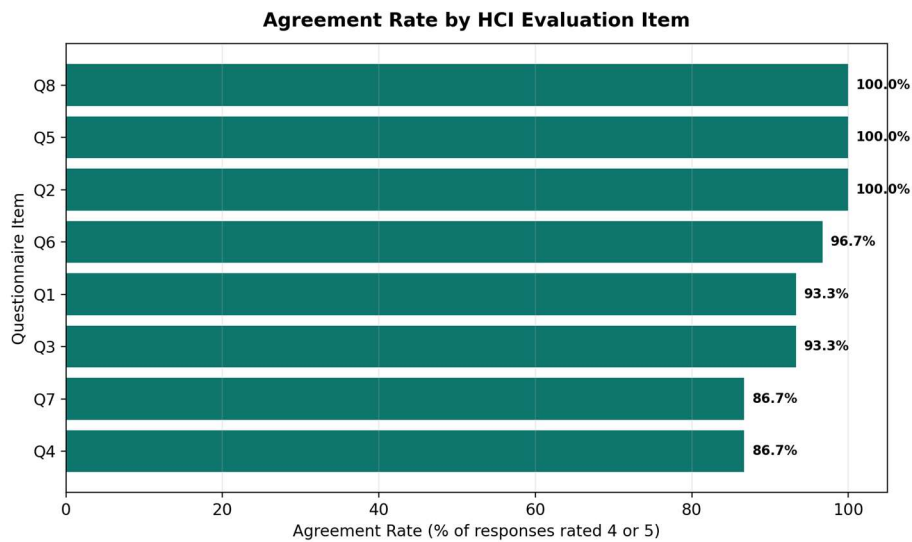


Figure 21. Illustrative agreement rate (responses rated 4 or 5) by evaluation item (synthetic demonstration data). Source: Authors' demonstration.